

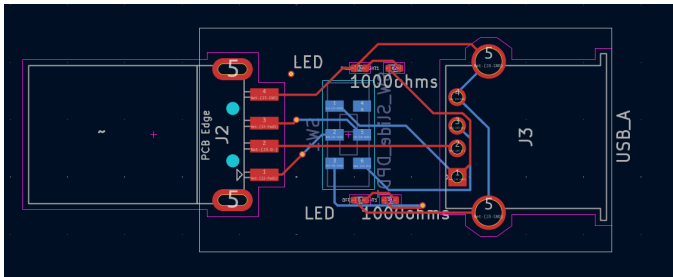
# USB-A Manual Switch Ver 1.0

## *A Physical USB Switch*

Project Report | 2026 | Haoyang Jiang

### 1. Project Overview

The primary goal of this project was to design and prototype a custom USB-A Manual Power Switch. This hardware module provides a physical "hard-cut" for USB-connected peripherals, addressing the need for enhanced privacy and convenient hardware resetting without the wear and tear of frequent plugging/unplugging.



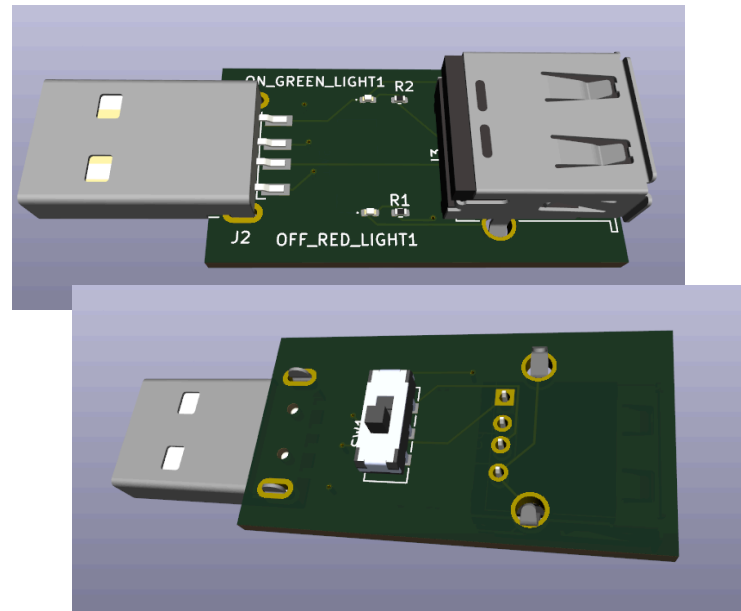
### Technical Scope

The development process involved a full-cycle Electronic Design Automation (EDA) workflow, encompassing:

- Schematic Capture: Designing the electrical logic and selecting appropriate components (USB connectors, LED indicators, and slide switches).
- PCB Layout: Translating the logic into a physical 2-layer board design using KiCAD 9.0.
- Mechanical Constraint Management: Managing height interference between high-profile USB connectors and low-profile switches.

## Key Features

- **Mixed-Signal Integration:** Successfully integrated Surface Mount Device (SMD) components for space-saving and Through-Hole Technology (THT) components for mechanical structural integrity.
- **Optimized 3-D Routing:** Utilized multi-layer routing strategies and via-interconnections to achieve a compact footprint.
- **Manufacturing Ready:** The final design passed all Design Rule Checks (DRC), meeting industrial standards for trace clearance, silk-screen alignment, and board edge integrity.



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## **2. Social and Practical Value**

### A. Absolute Security through Physical Disconnection

Unlike software-based toggles (which rely on the integrity of the Operating System and drivers), a physical switch provides an "Air-gapped" level of security.

- **Unhackable Nature:** Since the switch physically breaks the VCC (5V) or Data lines, no malware or remote exploit can bypass the hardware state. This is the only 100% reliable method to prevent unauthorized data exfiltration or peripheral hijacking.

### B. Psychology of Control: The "Sense of Agency"

Research in human-computer interaction suggests that users feel a significantly higher sense of agency (the feeling of being in control of one's actions and their effects) when using mechanical interfaces compared to digital ones.

- **Visibility vs. Obscurity:** A software icon can be spoofed or fail to update due to a system lag. A physical switch provides a permanent, visible state that requires no "trust" in the software—the user can see and feel the disconnected state.
- **The "Locus of Control":** Moving the privacy control from the "System" (External Locus) to the "User's hand" (Internal Locus) reduces "techno-anxiety." Users no longer need to wonder, *"Is the light really off because the camera is off, or because the hacker disabled the LED?"*

### C. Tactile Feedback and Closure (Psychological Satisfaction)

The use of a mechanical slide switch taps into the psychology of Haptic Feedback:

- **Sensory Confirmation:** The "click" or resistance of a physical switch provides immediate auditory and tactile confirmation. This sensory loop creates a psychological "Sense of Closure," delivering a level of satisfaction and "peace of mind" that a silent mouse click on a screen cannot replicate.
- **Ritualistic Security:** For high-stakes environments (e.g., sensitive meetings), the act of physically flicking a switch becomes a "security ritual" that reinforces the user's confidence in their environment's privacy.

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## **3. Challenges and Engineering Solutions**

### Challenge A: Vertical Geometric Interference

- The Problem: The standard USB-A Female connector (J3) has a significant vertical profile (approx. 7mm). When placed on the same side as the Slide Switch (SW1), the connector's casing physically obstructed finger access to the switch, making the device nearly impossible to operate.
- Engineering Solution: Implemented a Bottom-Layer Component Strategy. By flipping the switch to the Bottom Copper Layer (B.Cu) while keeping the connectors on the Top Layer (F.Cu), I utilized the PCB substrate as a physical separator. This maximized ergonomic accessibility without increasing the PCB's X-Y footprint.

### Challenge B: Cross-Layer Electrical Connectivity

- The Problem: Shifting the switch to the bottom layer disconnected it from the main 5V power bus on the top layer. I needed a way to route high-current power through the board without compromising signal integrity.
- Engineering Solution: I utilized Vias (Vertical Interconnect Access) to create "electrical bridges."
  - I routed the 5V input to a Via on the top layer, transitioned the signal to the bottom layer to interface with the switch, and then used another Via to bring the switched power back to the top-layer output.
  - Trace Width Optimization: To handle up to 2A of current, I increased the power trace width to 0.5mm, ensuring low resistance and preventing overheating.

### Challenge C: Mechanical Durability & Stress Management

- The Problem: USB ports are subject to frequent insertion and withdrawal forces, which can easily snap the solder joints of Surface Mount (SMD) connectors.
- Engineering Solution: I opted for Through-Hole Technology (THT) for the USB-A connectors. The THT pins penetrate the entire PCB and are soldered on both sides, providing

superior mechanical anchorage compared to SMD pads. This ensures the device can withstand long-term physical stress.

## Challenge D: Design Rule Constraint (DRC) Violations

- The Problem: During the final verification, the KiCAD DRC engine flagged "Silkscreen clipped by board edge" and "Malformed board outline" errors.
  - Engineering Solution: \* I performed a manual audit of the Edge.Cuts layer to ensure a perfectly closed-loop polygon.
    - I resolved the silkscreen warnings by modifying the J2 footprint, trimming the non-essential legend lines that extended beyond the manufacturing boundary. This ensured a clean, professional finish without any "bleeding" of ink during the fabrication process.
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## **4. Technical Summary**

This section summarizes the technical specifications and design standards achieved in the final version of the PCB.

### PCB Specifications

- Board Type: Dual-layer Rigid PCB (FR-4 Substrate).
- Dimensions: Compact form factor designed for portable USB integration.
- Copper Weight: 1 oz/ft<sup>2</sup> (Standard for stable power delivery).
- Surface Finish: HASL (Hot Air Solder Leveling) with Lead-Free compatibility.

### Electrical Parameters

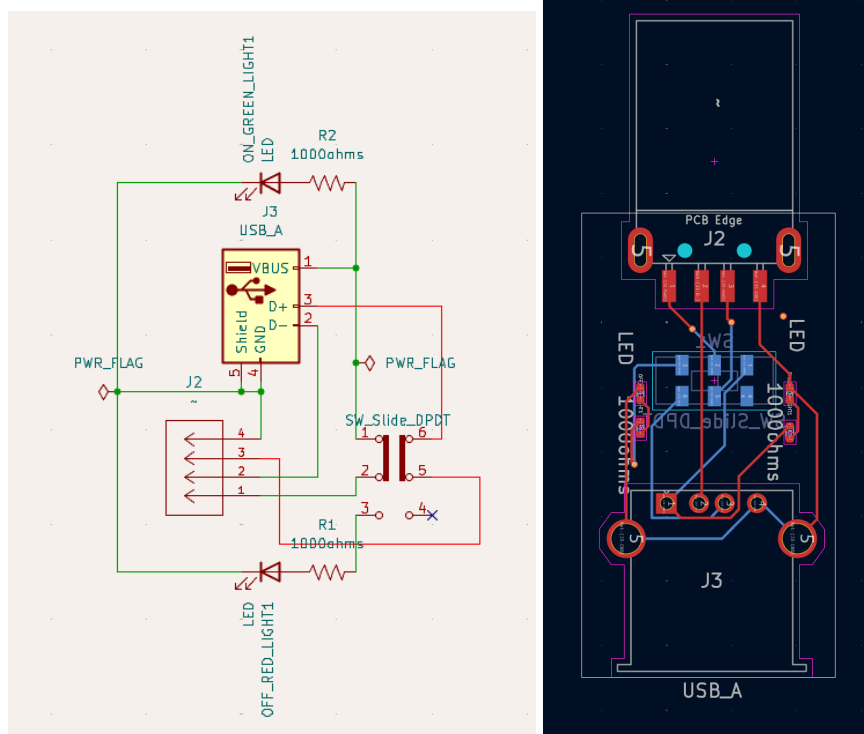
- Operating Voltage: 5.0V DC (Standard USB Bus Voltage).
- Current Capacity: Rated for up to 2.0A, supported by 0.5mm power traces to minimize voltage drop and thermal buildup.
- Signal Integrity: Maintains direct data line continuity (D+/D-) for peripheral recognition while switching the VCC line.

## Design Compliance (DRC Results)

The design was validated against industrial manufacturing constraints using the KiCAD 8.0 Design Rule Checker:

- Critical Errors: 0
- Unconnected Items: 0
- Minimum Clearance: 0.2 mm (Ensures no short circuits during fabrication).
- Minimum Trace Width: 0.25 mm (Signal) / 0.5 mm (Power).

## Schematic Overview:



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## **5. Relevance to Engineering Study**

This project served as a practical bridge between high school theoretical physics and applied engineering, reinforcing several core competencies required for undergraduate success:

### **A. Application of Theoretical Physics**

The design process required a deep understanding of Ohm's Law and Thermal Dynamics. Calculating the necessary trace width to support 2A of current without excessive heat generation transformed an abstract classroom formula into a critical design constraint. This reinforced the importance of using mathematical models to predict hardware behavior before fabrication.

### **B. Systems Thinking and Spatial Reasoning**

Working in a 3D CAD environment (KiCAD) demanded advanced spatial reasoning. Solving the interference between the USB-A housing and the switch through Bottom-Layer Implementation taught me how to view a system as a 3D entity rather than a 2D drawing. This aligns with the "Systems Thinking" approach central to Mechatronics and Electrical Engineering.

### **C. The Iterative Design Cycle**

The transition from a "Malformed Outline" error to a "0 Error / 0 Warning" state exemplified the Engineering Design Process (EDP). I learned that engineering is rarely about getting it right the first time; it is about systematic debugging, interpreting technical documentation, and iterating until the design meets all safety and manufacturing constraints.

## D. Attention to Detail and Documentation

The rigorous requirement for DRC (Design Rule Check) compliance taught me that precision is paramount. A single 0.2mm clearance violation could lead to a short circuit and project failure. This project cultivated a disciplined approach to technical documentation and rule-following that is essential for professional engineering practice.

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## **6. Conclusion & Future Work**

This project successfully demonstrates the feasibility of a high-current, compact USB power switch designed through advanced PCB routing techniques. By navigating complex 3-D spatial constraints and adhering to rigorous manufacturing standards, I have developed a hardware solution that bridges the gap between digital privacy needs and physical ergonomics. The successful elimination of all DRC errors confirms that the design is ready for industrial fabrication and practical deployment.

### Future Work

While the current version (v1.0) achieves its core functional goals, I intend to further iterate on the design in the following areas:

- **Enclosure Integration:** I plan to export the PCB as a STEP model and utilize Fusion 360 for collaborative mechanical design. This will involve creating a custom-fit, 3D-printed enclosure with integrated mounting bosses and precise port cutouts.
- **Smart Logic Expansion:** Future iterations could incorporate a microcontroller (e.g., an ATTiny85) to allow for timed power-offs or software-integrated logic, transitioning the project from a manual switch to a programmable power management hub.



- **Component Surface Optimization: Exploring ENIG (Electroless Nickel Immersion Gold)**  
surface finishing to improve long-term contact reliability and oxidation resistance for the USB terminals.

— *End of Report* —

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